

Salt restriction in hypertension—the effect of dietary advice and self monitoring of chloride concentration in urine

E. MELAND,* E. LAERUM† & R. J. ULVIK‡

*Department of Public Health and Primary Health Care, Division for General Practice, Ulriksdal 8c, 5009 Bergen, Norway, †Department of General Practice, Blindern, 0318 Oslo, Norway and

‡Laboratory of Clinical Biochemistry, Haukeland Universityhospital, 5021 Bergen, Norway

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The aims of this present study were firstly to examine whether diet compliance and blood pressure effect could be enhanced by self monitoring with a titrator strip (Quantab 1176) measuring the urine chloride concentration. This was achieved by an open, randomized parallel group design. The study also sought to evaluate the blood pressure effect of a moderately salt restricted diet by using a pre-test–post-test design with a run-in period and controlling for relevant confounding factors (weight, training and alcohol consumption). Furthermore, the study aims were to validate the measurement of chloride concentration in the morning urine by the Quantab titrator strip. We compared Quantab 1176 measurement of chloride concentration in the morning urine with 24 h sodium excretion, determined by the clinical chemical laboratory, was performed.

Twenty men and 14 women (mean age 53 years) with essential hypertension (mean: 165/96 mmHg) were observed during a run-in period of 4 weeks before randomization to either dietary advice combined with self monitoring of morning urine chloride concentration for 12 weeks, or dietary advice alone.

The reduction in diastolic blood pressure of 6 mmHg was not different in the two groups (between groups $p = 0.44$). Within group changes of systolic blood pressure were 10 mmHg and 6 mmHg ($p = 0.006$ and $p = 0.04$) in the diet plus Quantab group, and the diet only group respectively (between groups $p = 0.30$). No significant difference in 24 h sodium excretion could be detected between the groups. The morning urine chloride concentration correlated moderately to the 24 h urine sodium excretion ($r = 0.66$, $p < 0.001$).

It is concluded that self monitoring was not effective either to enhance the salt restriction diet compliance or the blood pressure effect in this study. A diastolic blood pressure reduction of 6 mmHg was attained by simple dietary advice.

Key words: controlled clinical trial, hypertension, self monitoring, sodium-restricted diet

Eivind Meland, Department of Public Health and Primary Health Care, Division for General Practice, Ulriksdal 8c, 5009 Bergen, Norway

Although the effect of salt restriction in hypertension is debated [1], a blood pressure (BP) lowering effect modified by age, initial BP and treatment duration, has been well documented [2, 3, 4]. However, as sodium chloride is such an important part of Western everyday diet, methods for enhancing compliance with long term diet change are called for [5].

Self monitoring is regarded as an important tool to induce behavioural change [6, 7]. Feedback from doctor to patient on measurement of total chloride excretion in urine may improve diet compliance [8], but this method is too comprehensive for self monitoring. A strip method for measuring urine chloride concentration may be more feasible.

The purpose of the present study was to examine whether diet compliance and blood pressure effect could be enhanced by self monitoring of chloride concentration in the morning urine by a titrator strip. The titrator strip was accordingly validated, and the effect of the dietary advice was evaluated.

MATERIAL AND METHOD

Setting and design

Patients were recruited from four general practice health centres during 1990, and examined in their respective centres by a GP and a specially trained technical assistant. An open, randomized parallel group design was chosen to evaluate the effect of self monitoring. The trial was designed with a 4 week run-in period, controlling for the regression effect. Other changes during the trial were also controlled for, allowing a pre-test–post-test evaluation of the blood pressure effect from a moderately salt restricted diet.

Patients

Patients were eligible if their diastolic (DBP) or systolic blood pressure (SBP) on both occasions in the run-in period exceeded the limits according to an age- and sex-dependent treatment programme [9] (male BP limits: 155/92 mmHg (30 years) to 172/107 mmHg (70 years); female BP limits: 165/92 mmHg to 200/107 mmHg).

Exclusion criteria included the use of anti-inflammatory drugs and the presence of

complicating heart and vessel disease demanding change or institution of additional hypotensive drugs during the trial. Patients with a DBP exceeding 115 mmHg were also excluded.

Twenty men and 14 women aged 29–70 y (mean age 53 y) were enrolled. As salt sensitivity depends on age [4, 10], patients were stratified in two age groups (≤ 56 , > 56). They were randomized to either diet treatment combined with self monitoring of morning urine chloride concentration or identical diet treatment alone. The participating GP gave oral information supported with written advice as listed below [11].

- 1 Fresh fish and meat should be preferred for dinner. Canned, salted or smoked food is only rarely allowed.
- 2 Fruit and vegetables should be used plentifully.
- 3 Boil potatoes without salt.
- 4 Salt should not be used during food processing. However, a little potassium-enriched salt may be added after food is prepared.
- 5 Spices and herbs should be used plentifully.
- 6 Lemon juice adds flavour to your food.
- 7 Roasting your food in the oven or microwave conserves its natural flavours.
- 8 When frying, use oil instead of butter.
- 9 Ask for salt-reduced soups or sauces if you are not preparing your own.
- 10 For baking bread, use 1 teaspoon salt per litre of liquid.

The self monitoring group measured their fasting morning urine chloride concentration on six different occasions during the trial. A recommendation of 30–50% reduction of urine chloride concentration compared with the initial value at the inclusion visit, was set.

Twelve of the patients were on hypotensive drug treatment; 9 patients were on monotherapy; 2 patients had duotherapy and one patient received four hypotensive drugs. The drugs used were ACE inhibitors (4); beta blockers (3); alpha blockers (2); alpha-methyl-dopamine (2); thiazide diuretics (2); aldosterone antagonist (2) and calcium channel blockers (2).

BP was measured after 3 minutes' rest in a sitting position using an semi-automatic, oscillometric monitor (Colin Pressmate BP-8800, Japan). The mean of three readings with intervals of 2 minutes was recorded. To monitor diet compliance, all patients collected 16 h and 8 h

urine specimens for measurement of electrolyte excretion at start, after 1 month and 3 months dietary treatment. Dietary advice was repeated at the one month visit.

Fasting high density lipoprotein (HDL) cholesterol was measured by a commercial standard procedure in the clinical chemical laboratory (HDL-cholesterol assay, Boehringer Mannheim GmbH, Diagnostica, Germany). The resting pulse was registered automatically at the last BP reading at each visit. Body weights were measured to control for weight change during the trial.

As indicators of alcohol consumption, glutamyltransferase (GT) and ferritin were measured [12] using standard procedures (GT: The Chem 1 Analytical System, Technicon Instr Corp, NY, USA; Ferritin: Fer-iron radio-immuno-assay, Ramco Laboratories Inc, Houston, Texas, USA). The coefficients of analytical variations were 2–3% for the GT and HDL-cholesterol methods and 7% for the ferritin assay. At the start and end of the trial patients recalled their alcohol consumption during the last month, using the items of alcohol intake from a validated questionnaire [13]. All patients gave their informed consent, and the trial was approved by the regional Ethical Committee.

Validation study

The fasting morning urine chloride concentration was determined with a chloride titrator strip (Quantab 1176) impregnated with silver dichromate. The length of the white colour change (silver chloride) is read on a numbered scale of the strip, and the concentration of chloride is determined from a calibration table. The concentrations of sodium and chloride in the urine were determined in an automated routine analyzer (Astra 8 system, Astra, Beckman Instruments Inc, CA, USA), and with a

coefficient of analytic variation between 1% and 2%. As two separate urine specimens 16 h (day) and 8 h (night) were taken, night urine chloride concentration was validated against 24 h chloride and sodium excretion.

Statistics

The trial was designed to reveal a DBP difference between groups of 5 mmHg with a significance level of 0.05 and power 0.9. Fourteen patients were required in each group presupposing a standard deviation of DBP change of 4.4 mmHg. To decide whether an observed within-group change of BP and sodium excretion was statistically significant, a one sided Wilcoxon's signed-rank test was used. When comparing mean changes between-groups, a one sided Wilcoxon's rank-sum test was chosen. In accordance with one sided testing, the 90% confidence interval (CI) was calculated. To evaluate possible confounders a two sided Wilcoxon's signed-rank test and the 95% CI were chosen. For validation of the Quantab titrator strip, the Pearson correlation coefficient and the 'limits of agreement' [14] were computed. The statistical package SPSS/PC program release 2.0 was used [15].

RESULTS

Clinical trial

The characteristics of the two study groups are given in Table I.

No change in DBP was observed during the run-in period, however, the SBP was reduced ($p = 0.09$ and $p = 0.008$ in the diet plus Quantab group and the diet only group respectively). Table II shows a DBP reduction of 6 mmHg in both groups during the trial ($p = 0.003$ and $p = 0.01$). SBP was also reduced

TABLE I. The characteristics of the two study groups at inclusion of the trial

Study group	Diet only group, n = 19, mean (95% CI*)	Diet plus Quantab group, n = 15, mean (95% CI*)
Age (years)	53 (48–58)	52 (47–57)
Body weight (kg)	80 (75–85)	81 (73–89)
Body Mass Index	27.1 (25.5–28.7)	26.8 (24.8–28.8)
Drug treated (%)	37 (13–61)	33 (6–60)
Gender (% females)	37 (13–61)	47 (18–75)

*Confidence interval.

TABLE II. Mean systolic and diastolic blood pressure before run-in period, at inclusion time and after 12 weeks of dietary treatment in 34 patients with hypertension

Treatment	Weeks	Systolic blood pressure, mmHg (90% CI)	Diastolic blood pressure, mmHg (90% CI)
Diet only group (n = 19)	-4	164 (157-171)	96 (94-100)
	0	157 (151-164)	95 (92-98)
	12	151 (145-156)	89 (86-93)
Diet plus Quantab group (n = 15)	-4	166 (156-175)	95 (91-99)
	0	160 (151-169)	95 (91-99)
	12	150 (142-157)	89 (85-93)
Within-group p-value*	Diet only group	0.04	0.01
	Diet plus Quantab group	0.006	0.003
Between-groups p-value**		0.30	0.44

*Change from intervention start to 12 weeks within each group.

**Difference in change between the groups.

in both groups (10 mmHg, $p = 0.006$ and 6 mmHg, $p = 0.04$), but no between-groups difference was observed.

As shown in Figure 1, a statistically significant reduction of 24 h urine sodium excretion was demonstrated in both groups after one and 3 months' dietary treatment. A mean reduction of 38 mmol Na 24 h^{-1} ($p = 0.03$) and 43 mmol Na 24 h^{-1} ($p = 0.01$) was obtained during 3 months, but no significant difference between the groups could be found. Potassium excretion increase was negligible during the trial (4 mmol

24 h^{-1} and 3 mmol 24 h^{-1} in the diet plus Quantab group and the diet only group respectively).

Possible confounders that could explain the blood pressure reduction are listed in Table III. No statistically significant changes were revealed with the exception of a significant reduction in serum HDL-cholesterol from 1.37 mmol l^{-1} to 1.22 mmol l^{-1} ($p = 0.004$). The individual changes of resting pulse, weight and alcohol consumption were nearly normally distributed, and no significant correlations between the individual changes of the 3 variables with change in DBP could be found.

Validation study

A strong correlation between chloride concentration measured by the hospital laboratory and by Quantab 1176 strip was found ($r = 0.97$, $n = 13$), but the limits of agreement were wide (23 to -24 mmol l^{-1}). A correlation coefficient of 0.97 ($n = 55$) between 24 h excretion of sodium and chloride was also found. Twenty-four hour sodium excretion correlated moderately with chloride concentration in the morning urine ($r = 0.66$, $p < 0.001$, $n = 49$), and correlated strongly with 8 h over-night chloride excretion ($r = 0.76$, $p < 0.001$, $n = 49$).

DISCUSSION

In this study self monitoring of urine chloride excretion did not improve dietary compliance or the BP lowering effect of salt restriction.

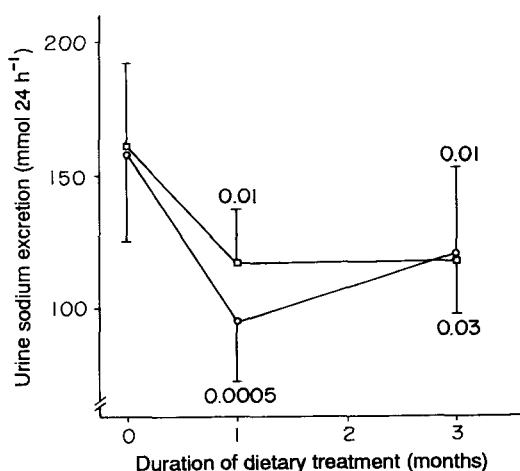


FIG. 1. Mean 24 h sodium excretion ($\text{mmol } 24\text{ h}^{-1}$) in the two study groups of hypertensive patients receiving a sodium restricted diet ($\square$ diet only group, $\circ$ diet plus Quantab group). Ninety percent confidence limits are indicated with p-values of within-group differences compared to initial values. Between-group p-values were 0.13 and 0.41 after one and 3 months respectively.

TABLE III. Possible confounding factors that might explain the blood pressure reduction during the intervention in 34 hypertensive patients

Parameter	At inclusion, mean (95% CI)	After 12 weeks, mean (95% CI)	p-value*
Weight (kg)	81 (76–85)	81 (76–85)	0.3
Resting puls (min ⁻¹)	72 (67–77)	70 (65–75)	0.08
HDL-cholesterol (mmol l ⁻¹)	1.37 (1.19–1.55)	1.22 (1.08–1.38)	0.004
Glutamyltransferase (U l ⁻¹)	34 (22–46)	40 (23–56)	0.3
Ferritin (μg l ⁻¹)	66 (37–95)	72 (41–103)	0.3
Alcohol units consumed last month**	12 (6–17)	18 (7–30)	0.4

*Change from start to 12 weeks within both groups.

**One unit equals 12.8 g ethanol.

The standard deviation of the DBP change was greater than assumed (6.6 mmHg as opposed to 4.4 mmHg), leading to a reduced power of the study. Even with an actual power in the range of 0.7–0.8 the negative conclusion seems tenable. A DBP reduction of 6 mmHg was attained by 40 mmol 24 h⁻¹ decline of mean sodium excretion irrespective of self monitoring.

The validation study supports the previous observation [8] that sodium and chloride excretion are strongly correlated. Although urine chloride concentration assessed with the Quantab titrator strip and by standard method are strongly correlated, the wide limits of agreement are unacceptable for clinical purposes.

Kaplan *et al.* [8] demonstrated an effect on BP from feedback on over-night total chloride excretion estimated by Quantab strip. It is, however, unclear whether the control group had the same dietary advice and follow-up as the feedback group during their trial. Self monitoring is held to be effective in behavioural techniques [6, 7], however, empirical data are conflicting [16].

The reduction of sodium excretion demonstrated in this study is in line with other similar studies [3, 17, 18]. A reduction of sodium excretion corresponding to 40–50 mmol d⁻¹ is attainable by simple dietary advice. The BP effect in this study slightly exceeds the expected effect from moderate salt restriction [3, 18]. Caution must be used, however, when interpreting within-group changes where no significant differences between groups are found. Therefore we controlled for possible confounders such as the regression effect and changes of weight, exercise and alcohol consumption [1].

The DBP was stable during the run-in period; the SBP was reduced. Regression effect might

therefore partly explain the SBP reduction demonstrated during the trial. The decreased HDL-cholesterol level and the stable resting pulse make changes of exercise improbable [19, 20]. No weight reduction was found during the trial. Ferritin and GT are accepted biochemical markers of alcohol consumption especially when measuring changes over time [12]. These markers and the anamnestic questionnaire could not reveal any significant change in alcohol consumption. In addition, individual changes of body weight, resting pulse and alcohol consumption correlated insignificantly with DBP changes. Therefore, the importance of these confounding factors seems trivial.

The BP reduction might have been caused by hypotensives, if drug compliance tended to improve during the trial. However, there was no significant difference between drug treated and untreated (DBP reduced by 7 mmHg and 5 mmHg, $p = 0.5$). Thus, although regression may partly explain the SBP reduction, it is likely that the reduced DBP in this study resulted from salt restriction.

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REFERENCES

- 1 The Subcommittee on Nonpharmacological Therapy of the 1984 Joint National Committee on Detection, Evaluation and Treatment of High Blood Pressure. Nonpharmacological approaches to the control of high blood pressure. *Hypertension* 1986; 8: 444–67.
- 2 Grobbee DE, Hofman A. Does sodium restriction lower blood pressure? *BMJ* 1986; 293: 27–9.
- 3 Law MR, Frost CD, Wald NJ. III—Analysis of data from trials of salt reduction. *BMJ* 1991; 302: 819–24.
- 4 Myers JB. Reduced sodium chloride intake normalises blood pressure distribution. *J Hum Hypertens* 1989; 3: 97–104.
- 5 Nugent CA, Carnahan JE, Sheehan ET, Myers C. Salt restriction in hypertensive patients. Comparison of advice, education, and group management. *Arch Intern Med* 1984; 144: 1415–7.
- 6 Fraser GE. Preventive cardiology. New York: Oxford University Press 1986: 319–45.
- 7 Bandura A. Social learning theory. Englewood Cliffs, Prentice Hall, Inc. 1977: 160–91.
- 8 Kaplan NM, Simmons M, McPhee C, Carnegie A, Stefanu C, Cade S. Two techniques to improve adherence to dietary sodium restriction in the treatment of hypertension. *Arch Intern Med* 1982; 142: 1638–41.
- 9 Holmen J, Forsdahl A, Kvamme J-I, Stokke HP. High blood pressure. NSAMs program on management in general practice. Oslo: GHE-K/report No 11, 1986.
- 10 Intersalt Cooperative Research Group. Intersalt: an international study of electrolyte excretion and blood pressure. Results for 24 hour urinary sodium and potassium excretion. *BMJ* 1988; 297: 319–28.
- 11 Langford HG, Schlundt D, Levine K. Sodium restriction in hypertension. *Comp Ther* 1984; 10: 6–11.
- 12 Lundin L, Hällgren R, Birgegård G, Wide L. Serum ferritin in alcoholics and the relation to liver damage, iron state and erythropoietic activity. *Acta Med Scand* 1981; 209: 327–31.
- 13 Saunders JB, Aasland OG, Amundsen A, Grant M. Alcohol consumption and related problems among primary health care patients: WHO Collaborative Project on Early Detection of Persons with Harmful Alcohol Consumption—I. *Addiction* 1993; 88: 349–62.
- 14 Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet* 1986; I: 307–10.
- 15 Norusis MJ. SPSS/PC+ V2.0. Base manual. Chicago: SPSS Inc, 1988; B190–2.
- 16 Wysocki T. Impact of blood glucose monitoring on diabetic control: obstacles and interventions. *J Behav Med* 1989; 12: 183–205.
- 17 Stamler R, Stamler J, Grimm R *et al.* Nutritional therapy for high blood pressure. *JAMA* 1987; 257: 1484–92.
- 18 Australian National Health and Medical Research Council Dietary Salt Study Management Committee. Fall in blood pressure with modest reduction in dietary salt intake in mild hypertension. *The Lancet* 1989; 1: 399–402.
- 19 Brownell KD, Bachorik PS, Ayerle RS. Changes in plasma lipid and lipoprotein levels in men and women after a program of moderate exercise. *Circulation* 1982; 3: 477–84.
- 20 Wood PD, Stefanick ML, Dreon DM *et al.* Changes in plasma lipids and lipoproteins in overweight men during weight loss through dieting as compared with exercise. *N Engl J Med* 1988; 319: 1173–9.

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